

Course name: Models of the motor system 2026-2
Course number: 367-1-4431

Lecturer: Opher Donchin

Syllabus: This course introduces biomechanical and neural modeling approaches, with a focus on how they apply to human movement and motor control. Students will explore key engineering models of the motor system, emphasizing both the sensory and motor components, and how these models explain the complexity of movement and learning. The complexity of the various aspects of the motor system will be addressed by considering different specific aspects and how they are modelled. Specifically, we will cover sensory systems and multisensory integration through the Bayesian perspective; we will consider the complexity of muscles and muscle models and how they can be simplified with an approach focusing on kinematics and dynamics. Then, we will look at how the system learns and consider different learning algorithms. Finally, we will look at multi-state models that bring together the different aspects into a fuller framework.

By the end of the course, students will be able to:

- Implement biomechanical and neural models of the motor system using Python.
- Analyze and interpret sensory and motor control models in human movement.
- Apply Bayesian principles to multisensory integration and learning in motor systems.
- Critically evaluate the strengths and limitations of different modeling approaches.

The course balances conceptual understanding with practical modeling skills, ensuring students can both interpret and implement motor system models.

Programming for the course will be in Python. Each meeting will consist of 90 minutes of lecture followed by an in-class modeling assignment for students to complete during the second half of lecture (submitting the in-class assignments will lead to up to 5% bonus credit). Additionally, there will be 4 programming-based homeworks over the course of the semester.

There will be a final exam. The exam will be 2 hours with multiple choice questions. It will primarily cover material in the lectures and the required papers (covered by the lectures).

Communication with the instructors: by e-mail

Office hours: by appointment

Assessment Breakdown:

- **Homework Assignments (3 total, 50%):** Each assignment requires students to implement and analyze specific motor system models in Python. Late submission of exercises will lead to a 4% penalty in the first week after the due date, an 8% penalty in the second week, and a 12% penalty after that.
- **Final Project (35%):** Reading and analyzing a paper on motor control that includes a modeling component. The grade will be based on written work, but a

short, in person, evaluation will be required in order to receive full credit for the assignment. A student whose in person evaluation suggests that they did not actually understand the paper, the model, or their own answers will be allowed to redo the assignment on a new paper for a maximum of 70% credit.

- **In-class modeling (15%):** The last hour of each class will be devoted to work on an in-class assignment covering the material. Students submitting a working solution in class will receive full credit. 90% credit can be earned by submitting a working solution before the next class meeting. 60% credit can be earned by submitting a working solution before the end of the semester.

To pass the course you must pass the exam and submit all the exercises, and get a passing grade in both

Lecture #	Date	Topics	Comments
1	17.3	Introduction to the motor system	
2	24.3	Muscles and muscle models	Homework 1 handed out
	7.4 (Passover)	<i>No class</i>	
3	14.4	Equations for kinematics and dynamics	
	21.4 (Memorial day)	<i>No class</i>	
	28.5		
4	5.5	Sensory systems	
5	12.5	Multi-sensory integration, Bayesian estimation	Homework 1 due
6	19.5	Motor "laws" and basic kinematics	
7	26.5 (Id El Adha)	Internal models and error learning	Homework 2 handed out
8	3.6 (Student's day)	Reward learning	
9	9.6	Multi-state model of motor learning	Homework 2 due Homework 3 handed out
10	16.6	Combining multiple types of learning	
11	23.6	Behavioral and neural manifolds	Homework 3 due
12	30.6	Project discussion	